

**Цель работы:** сформировать умение работать с Protocol Data Unit, используя программу Wireshark.

## Выполнение заданий: Вариант 11

### 1. Отфильтровать отправленные пакеты

Wireshark capture showing filtered outgoing packets (ip.dst == 192.168.0.119). The packet list shows various TCP and TLSv1.2 packets. The packet details pane shows the structure of a TCP segment. The packet bytes pane shows the raw data.

| No.   | Time        | Source        | Destination   | Protocol | Length | Info                                                                                       |
|-------|-------------|---------------|---------------|----------|--------|--------------------------------------------------------------------------------------------|
| 18647 | 1576.372526 | 52.109.77.1   | 192.168.0.119 | TCP      | 1514   | 443 → 54812 [ACK] Seq=1461 Ack=216 Win=525312 Len=1460 [TCP segment of a reassembled PDU]  |
| 18649 | 1576.372787 | 52.109.77.1   | 192.168.0.119 | TCP      | 1514   | 443 → 54812 [ACK] Seq=2921 Ack=216 Win=525312 Len=1460 [TCP segment of a reassembled PDU]  |
| 18650 | 1576.373147 | 52.109.77.1   | 192.168.0.119 | TLSv1.2  | 1484   | Server Hello, Certificate, Certificate Status, Server Key Exchange, Server Hello Done      |
| 18653 | 1576.440779 | 52.109.77.1   | 192.168.0.119 | TLSv1.2  | 105    | Change Cipher Spec, Encrypted Handshake Message                                            |
| 18654 | 1576.440779 | 52.109.77.1   | 192.168.0.119 | TLSv1.2  | 123    | Application Data                                                                           |
| 18659 | 1576.500887 | 52.109.77.1   | 192.168.0.119 | TLSv1.2  | 92     | Application Data                                                                           |
| 18660 | 1576.505875 | 52.109.77.1   | 192.168.0.119 | TLSv1.2  | 306    | Application Data                                                                           |
| 18662 | 1576.550813 | 52.109.77.1   | 192.168.0.119 | TCP      | 54     | 443 → 54812 [ACK] Seq=6221 Ack=929 Win=524544 Len=0                                        |
| 18675 | 1578.686287 | 37.214.76.89  | 192.168.0.119 | TCP      | 74     | 19036 → 54628 [PSH, ACK] Seq=40305 Ack=47745 Win=29340 Len=20                              |
| 18676 | 1578.715693 | 37.214.76.89  | 192.168.0.119 | TCP      | 54     | 19036 → 54628 [ACK] Seq=40405 Ack=47785 Win=30720 Len=0                                    |
| 18679 | 1578.716122 | 37.214.76.89  | 192.168.0.119 | TCP      | 74     | 19036 → 54628 [PSH, ACK] Seq=40405 Ack=47785 Win=30720 Len=20                              |
| 18685 | 1581.520990 | 46.53.242.153 | 192.168.0.119 | TCP      | 74     | 14992 → 54628 [PSH, ACK] Seq=17057 Ack=5358 Win=30232 Len=20                               |
| 18688 | 1581.850388 | 46.53.242.153 | 192.168.0.119 | TCP      | 74     | [TCP Spurious Retransmission] 14992 → 54628 [PSH, ACK] Seq=17057 Ack=5358 Win=30232 Len=20 |
| 18690 | 1582.216538 | 46.53.242.153 | 192.168.0.119 | TCP      | 74     | [TCP Spurious Retransmission] 14992 → 54628 [PSH, ACK] Seq=17057 Ack=5358 Win=30232 Len=20 |
| 18692 | 1582.872511 | 46.53.253.30  | 192.168.0.119 | TCP      | 250    | 24420 → 54631 [PSH, ACK] Seq=58309 Ack=60565 Win=30436 Len=196                             |
| 18694 | 1582.946283 | 46.53.242.153 | 192.168.0.119 | TCP      | 74     | [TCP Spurious Retransmission] 14992 → 54628 [PSH, ACK] Seq=17057 Ack=5358 Win=30232 Len=20 |
| 18696 | 1583.024318 | 46.53.242.153 | 192.168.0.119 | TCP      | 54     | 14992 → 54628 [ACK] Seq=17077 Ack=5398 Win=30192 Len=0                                     |
| 18697 | 1583.027544 | 46.53.242.153 | 192.168.0.119 | TCP      | 74     | 14992 → 54628 [PSH, ACK] Seq=17077 Ack=5398 Win=30192 Len=20                               |
| 18698 | 1583.040864 | 46.53.242.153 | 192.168.0.119 | TCP      | 74     | [TCP Retransmission] 14992 → 54628 [PSH, ACK] Seq=17077 Ack=5398 Win=30192 Len=20          |

Frame 6791: 54 bytes on wire (432 bits), 54 bytes captured (432 bits) on interface \Device\NPF\_{C9BC9A2F-08B4-4720-95BA-CC645058EB3A}, id 0  
Ethernet II, Src: Tp-LinkT\_bd1a:64 (d8:47:32:bd:1a:64), Dst: IntelCor\_1c:87:bb (84:a6:c8:1c:87:bb)  
Internet Protocol Version 4, Src: 213.180.193.234, Dst: 192.168.0.119  
Transmission Control Protocol, Src Port: 443, Dst Port: 54708, Seq: 5311, Ack: 1897, Len: 0

0000 84 a6 c8 1c 87 bb d8 47 32 bd 1a 64 08 00 45 00 .....G 2...E-  
0010 00 28 96 a4 40 00 37 06 54 6d d5 b4 c1 ea c0 a8 .....@ 7 Tm.....  
0020 00 77 b1 bb d5 b4 b5 9a 74 85 29 32 76 9a 50 14 .....w.....t.)2v.P-  
0030 80 a6 b5 0f 00 00 .....  
Transmission Control Protocol (tcp), 20 byte(s) | Пакеты: 18699 · Показаны: 8036 (43.0%) | Профиль: Default

### 2. Отфильтровать отправленные и полученные пакеты

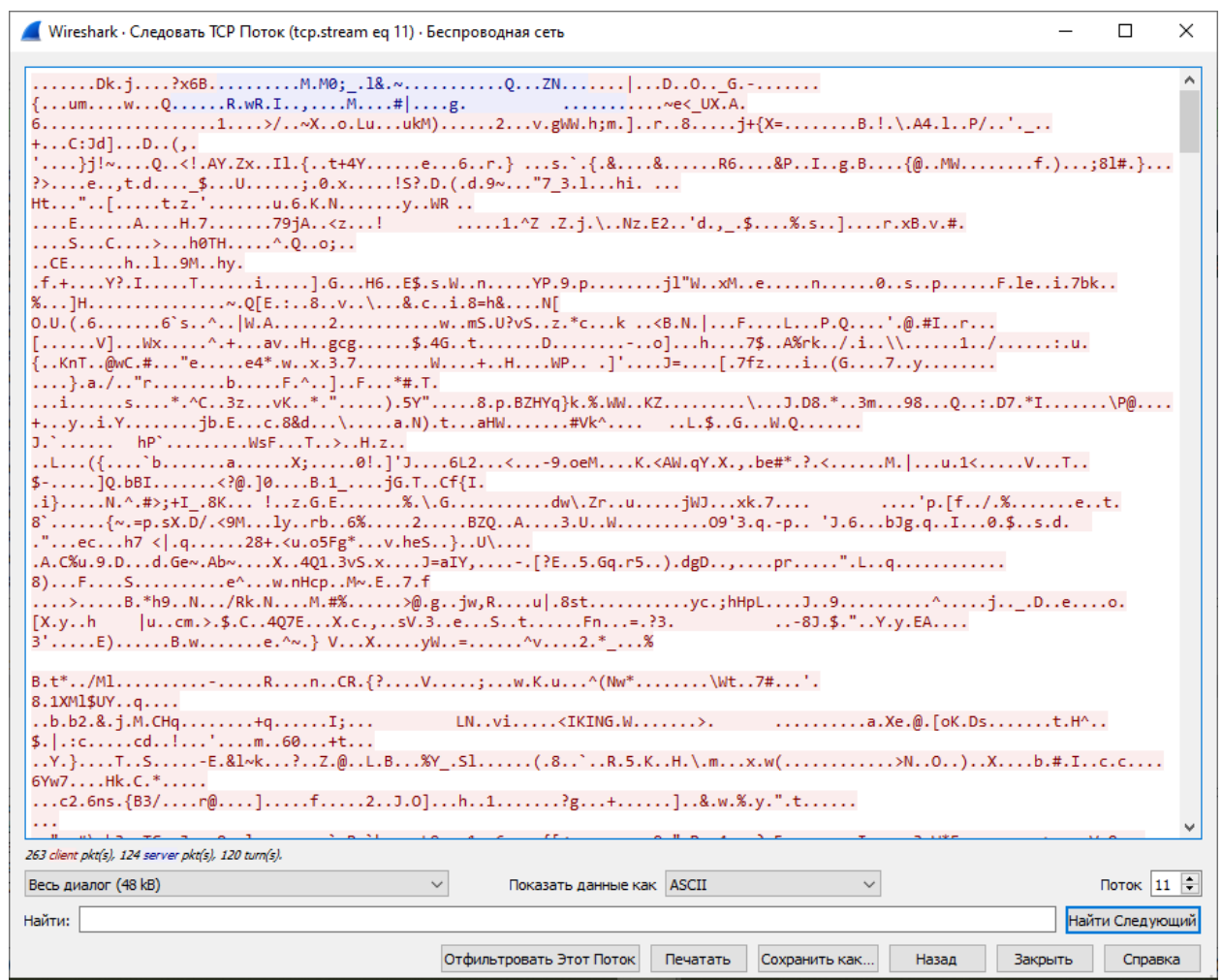
Wireshark capture showing filtered outgoing and incoming packets (ip.dst == 192.168.0.119 || ip.src == 192.168.0.119). The packet list shows various TCP and TLSv1.2 packets. The packet details pane shows the structure of a TCP segment. The packet bytes pane shows the raw data.

| No.   | Time        | Source        | Destination   | Protocol | Length | Info                                                          |
|-------|-------------|---------------|---------------|----------|--------|---------------------------------------------------------------|
| 18946 | 1632.292580 | 37.215.28.140 | 192.168.0.119 | TCP      | 74     | 1255 → 54628 [PSH, ACK] Seq=123613 Ack=8270 Win=30376 Len=20  |
| 18947 | 1632.292894 | 37.215.28.140 | 192.168.0.119 | TCP      | 74     | 1255 → 54628 [PSH, ACK] Seq=123633 Ack=8270 Win=30376 Len=20  |
| 18948 | 1632.292962 | 192.168.0.119 | 37.215.28.140 | TCP      | 54     | 54628 → 1255 [ACK] Seq=8270 Ack=123653 Win=29516 Len=0        |
| 18949 | 1632.293177 | 192.168.0.119 | 37.215.28.140 | TCP      | 74     | 54628 → 1255 [PSH, ACK] Seq=8270 Ack=123653 Win=29516 Len=20  |
| 18950 | 1632.356666 | 37.215.28.140 | 192.168.0.119 | TCP      | 54     | 1255 → 54628 [ACK] Seq=123653 Ack=8290 Win=30356 Len=0        |
| 18951 | 1632.448723 | 37.215.28.140 | 192.168.0.119 | TCP      | 298    | 1255 → 54628 [PSH, ACK] Seq=123653 Ack=8290 Win=30356 Len=244 |
| 18952 | 1632.502230 | 192.168.0.119 | 37.215.28.140 | TCP      | 54     | 54628 → 1255 [ACK] Seq=8290 Ack=123897 Win=30720 Len=0        |
| 18953 | 1633.461482 | 37.215.28.140 | 192.168.0.119 | TCP      | 298    | 1255 → 54628 [PSH, ACK] Seq=123897 Ack=8290 Win=30356 Len=244 |
| 18954 | 1633.502217 | 192.168.0.119 | 37.215.28.140 | TCP      | 54     | 54628 → 1255 [ACK] Seq=8290 Ack=124141 Win=30476 Len=0        |
| 18955 | 1634.475966 | 37.215.28.140 | 192.168.0.119 | TCP      | 298    | 1255 → 54628 [PSH, ACK] Seq=124141 Ack=8290 Win=30356 Len=244 |
| 18960 | 1634.517803 | 192.168.0.119 | 37.215.28.140 | TCP      | 54     | 54628 → 1255 [ACK] Seq=8290 Ack=124385 Win=30232 Len=0        |
| 18961 | 1635.484018 | 37.215.28.140 | 192.168.0.119 | TCP      | 298    | 1255 → 54628 [PSH, ACK] Seq=124385 Ack=8290 Win=30356 Len=244 |
| 18962 | 1635.533378 | 192.168.0.119 | 37.215.28.140 | TCP      | 54     | 54628 → 1255 [ACK] Seq=8290 Ack=124629 Win=29988 Len=0        |
| 18963 | 1638.689548 | 37.214.76.89  | 192.168.0.119 | TCP      | 74     | 19036 → 54628 [PSH, ACK] Seq=40945 Ack=47825 Win=30680 Len=20 |
| 18964 | 1638.689848 | 192.168.0.119 | 37.214.76.89  | TCP      | 74     | 54628 → 19036 [PSH, ACK] Seq=47825 Ack=40965 Win=29496 Len=20 |
| 18965 | 1638.690065 | 192.168.0.119 | 37.214.76.89  | TCP      | 74     | 54628 → 19036 [PSH, ACK] Seq=47845 Ack=40965 Win=29496 Len=20 |
| 18966 | 1638.711515 | 37.214.76.89  | 192.168.0.119 | TCP      | 54     | 19036 → 54628 [ACK] Seq=40965 Ack=47865 Win=30640 Len=0       |
| 18967 | 1638.714221 | 37.214.76.89  | 192.168.0.119 | TCP      | 74     | 19036 → 54628 [PSH, ACK] Seq=40965 Ack=47865 Win=30640 Len=20 |
| 18968 | 1638.767707 | 192.168.0.119 | 37.214.76.89  | TCP      | 54     | 54628 → 19036 [ACK] Seq=47865 Ack=40985 Win=29476 Len=0       |

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Ethernet II, Src: Tp-LinkT\_bd1a:64 (d8:47:32:bd:1a:64), Dst: IntelCor\_1c:87:bb (84:a6:c8:1c:87:bb)  
Internet Protocol Version 4, Src: 213.180.193.234, Dst: 192.168.0.119  
Transmission Control Protocol, Src Port: 443, Dst Port: 54708, Seq: 5311, Ack: 1897, Len: 0

0000 84 a6 c8 1c 87 bb d8 47 32 bd 1a 64 08 00 45 00 .....G 2...E-  
0010 00 28 96 a4 40 00 37 06 54 6d d5 b4 c1 ea c0 a8 .....@ 7 Tm.....  
0020 00 77 b1 bb d5 b4 b5 9a 74 85 29 32 76 9a 50 14 .....w.....t.)2v.P-  
0030 80 a6 b5 0f 00 00 .....  
Transmission Control Protocol (tcp), 20 byte(s) | Пакеты: 18968 · Показаны: 16362 (86.3%) | Профиль: Default

### 3. Просмотр потока TCP



#### 4. Просмотр неполадок в сети

Wireshark · Информация Эксперта · Беспроводная сеть

| Степень тяжести | Сводка                                                          | Группа    | Протокол  | Подсчёт |
|-----------------|-----------------------------------------------------------------|-----------|-----------|---------|
| > Error         | Malformed Packet (Exception occurred)                           | Malformed | SMTP      | 4       |
| > Error         | New fragment overlaps old data (retransmission?)                | Malformed | TCP       | 9       |
| > Warning       | DNS response retransmission. Original response in frame 9...    | Protocol  | DNS       | 3       |
| > Warning       | DNS query retransmission. Original request in frame 9141        | Protocol  | DNS       | 3       |
| > Warning       | DNS response retransmission. Original response in frame 2...    | Protocol  | mDNS      | 350     |
| > Warning       | DNS query retransmission. Original request in frame 882         | Protocol  | mDNS      | 236     |
| > Warning       | Ignored Unknown Record                                          | Protocol  | TLS       | 8       |
| > Warning       | TCP Zero Window segment                                         | Sequence  | TCP       | 3       |
| > Warning       | Connection reset (RST)                                          | Sequence  | TCP       | 38      |
| > Warning       | This frame is a (suspected) out-of-order segment                | Sequence  | TCP       | 33      |
| > Warning       | Previous segment(s) not captured (common at capture sta...      | Sequence  | TCP       | 37      |
| > Warning       | ACKed segment that wasn't captured (common at capture...        | Sequence  | TCP       | 5       |
| > Warning       | D-SACK Sequence                                                 | Sequence  | TCP       | 159     |
| > Note          | This frame is a (suspected) fast retransmission                 | Sequence  | TCP       | 4       |
| > Note          | A new tcp session is started with the same ports as an earli... | Sequence  | TCP       | 5       |
| > Note          | Didn't find padding of zeros, and an undecoded trailer exis...  | Protocol  | Ethertype | 11      |
| > Note          | This session reuses previously negotiated keys (Session res...  | Sequence  | TLS       | 7       |
| > Note          | ACK to a TCP keep-alive segment                                 | Sequence  | TCP       | 64      |
| > Note          | TCP keep-alive segment                                          | Sequence  | TCP       | 67      |
| > Note          | "Time To Live" != 255 for a packet sent to the Local Networ...  | Sequence  | IPv4      | 20      |
| > Note          | This frame is a (suspected) spurious retransmission             | Sequence  | TCP       | 42      |
| > Note          | Duplicate ACK (#1)                                              | Sequence  | TCP       | 234     |
| > Note          | This frame undergoes the connection closing                     | Sequence  | TCP       | 80      |
| > Note          | This frame initiates the connection closing                     | Sequence  | TCP       | 81      |
| > Note          | This frame is a (suspected) retransmission                      | Sequence  | TCP       | 134     |
| > Chat          | GET / HTTP/1.1\r\n                                              | Sequence  | HTTP      | 12      |
| > Chat          | M-SEARCH * HTTP/1.1\r\n                                         | Sequence  | SSDP      | 433     |
| > Chat          | Connection finish (FIN)                                         | Sequence  | TCP       | 161     |
| > Chat          | Connection establish acknowledge (SYN+ACK): server por...       | Sequence  | TCP       | 100     |
| > Chat          | Connection establish request (SYN): server port 443             | Sequence  | TCP       | 107     |

Дисплейный фильтр: "tcp.stream eq 11"

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## Контрольные вопросы

### 1. Что такое PDU?

Данные протокола unit (PDU) - это единичный блок информации, передаваемый между одноранговыми объектами компьютерной сети.

### 2. Назовите единицы PDU на 1-4 уровнях

Функции или услуги сети реализованы на отдельных «уровнях». Например, отправка единиц и нулей по проводу, оптоволокну и т. д. выполняется на физическом уровне, а организация единиц и нулей в блоки данных и их безопасная доставка в нужное место в сети осуществляется уровнем канала передачи данных, Передача блоков данных по нескольким подключенным сетям осуществляется сетевым уровнем, а доставка данных в нужное программное приложение в пункт назначения осуществляется транспортным уровнем.

### 3. Что делает протокол TCP?

Протокол управления передачей (TCP) реализует режим передачи с установлением соединения

4. Что называется ячейкой?

PDU уровня канала данных в асинхронном режиме передачи (ATM) сети называется ячейкой.

**Вывод:** я научился работать с Protocol Data Unit, используя программу Wireshark.